Skyhook

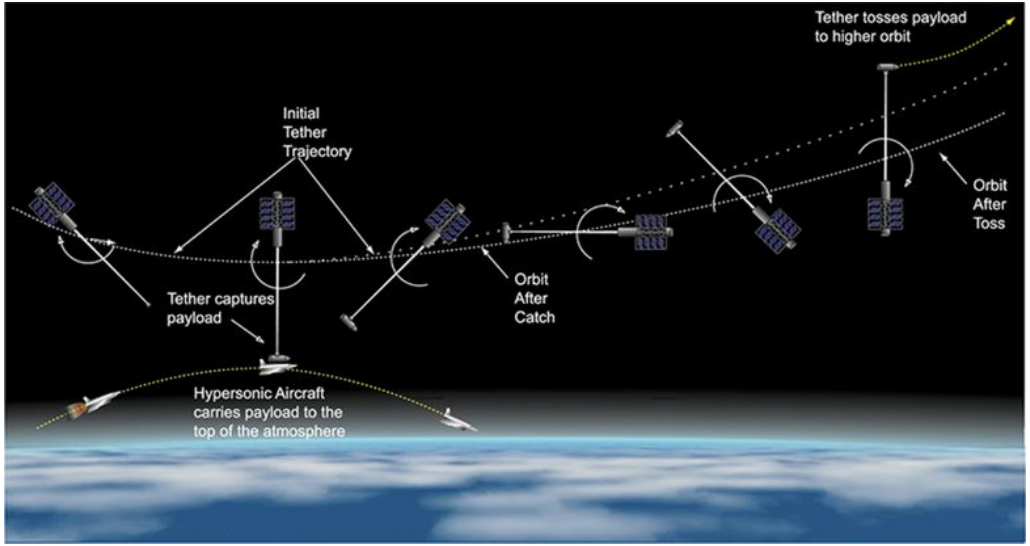
ME5060: Spacecraft Dynamics and Control

Yashwanth M
Harishankar M
Ashwin Ganesh

Dr. Vishnu R Unni

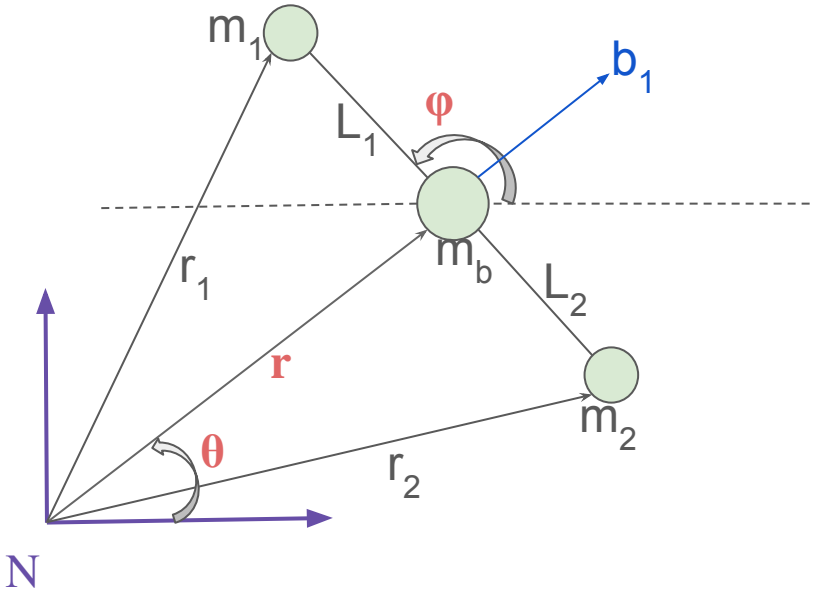
Getting to space is hard!: Skyhook for the save?

https://www.theguardian.com/science/2017/jun/08/india-launches-powerful-new-rocket-spacewatch



https://www.jalopnik.com/the-skyhook-could-slingshot-mankind-across-the-solar-sy-1851117250/

Dumbbell Satellite



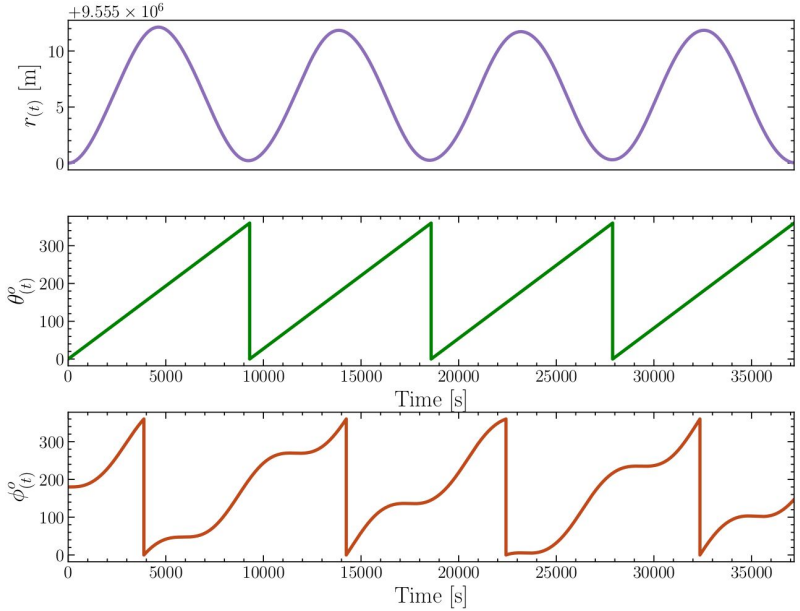
$$\ddot{r} = r\dot{\theta}^2 - \frac{GM_E}{m_1 + m_2 + m_b}\left[D_1 \cos(\varphi - \theta) + D_2 + \frac{m_b}{r^2}\right] \tag{6}$$

$$\ddot{\theta} = \frac{1}{r}\left[-2\dot{r}\dot{\theta} + \frac{GM_E}{m_1 + m_2 + m_b}\left[\frac{m_1 L_1}{r_1^3} - \frac{m_2 L_2}{r_2^3}\right]\sin(\varphi - \theta)\right] \tag{7}$$

$$\ddot{\varphi} = \frac{GM_E}{I}\left[\left[\frac{m_1 L_1}{r_1^3} - \frac{m_2 L_2}{r_2^3}\right] r \sin(\varphi - \theta)\right] \tag{8}$$

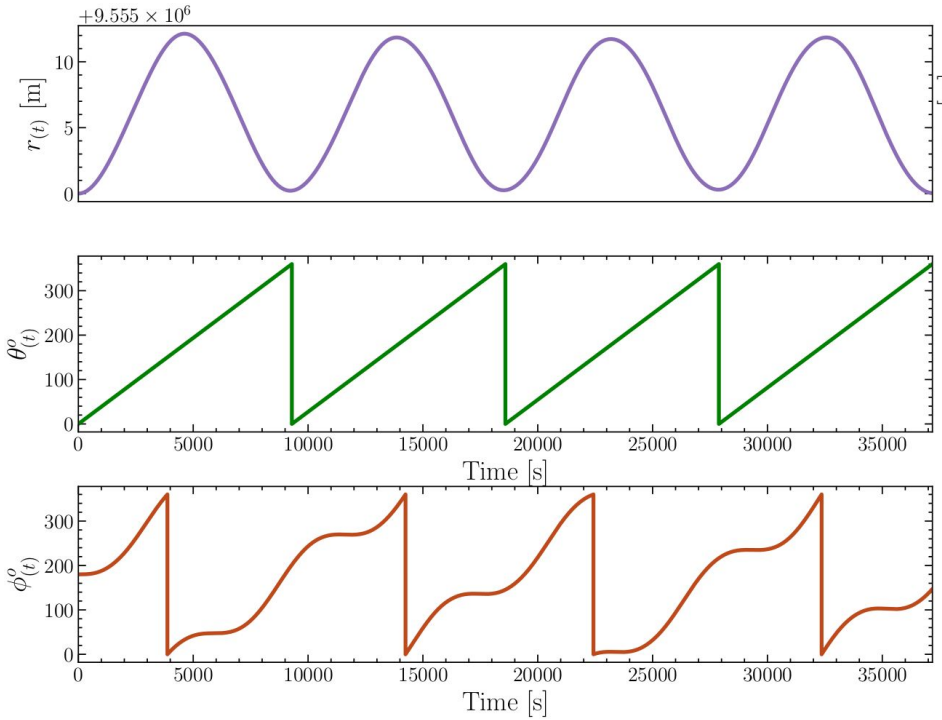
Parameter	Values
m_1	500 kg
m_2	500 kg
L_1	10 m
L_2	10 m
$r_{(0)}$	$1.5R_E$
$\varphi_{(0)}$	$\left(\frac{\pi}{2}\right)$
$\dot{\varphi}_{(0)}$	$1000\,\dot{\theta}_{(0)}$

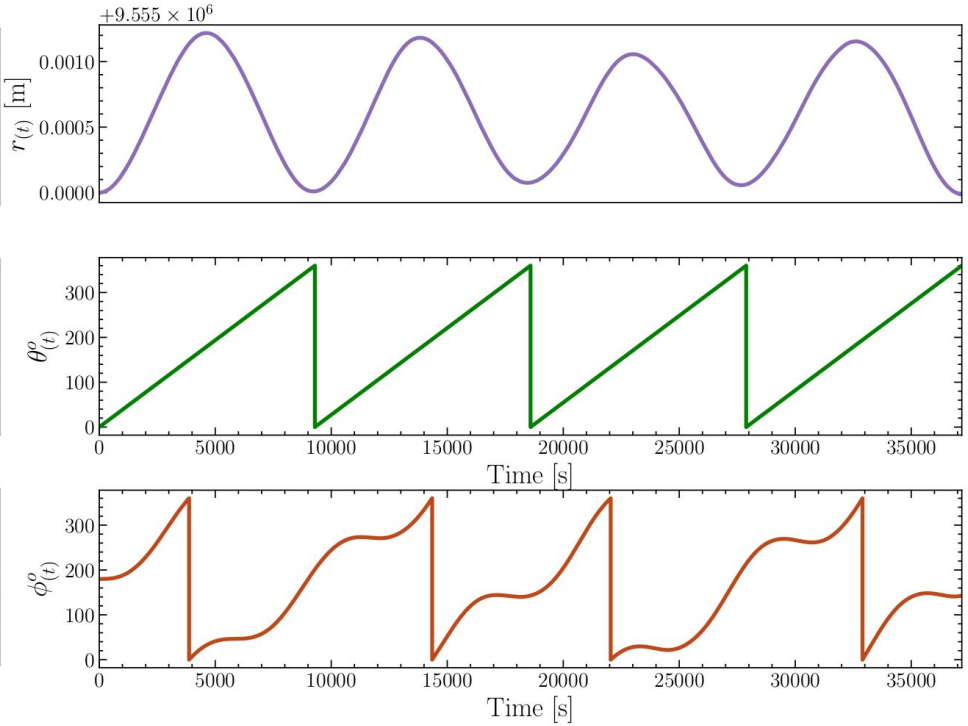
“Stability” of Dumbbell Satellite: Make it spin!



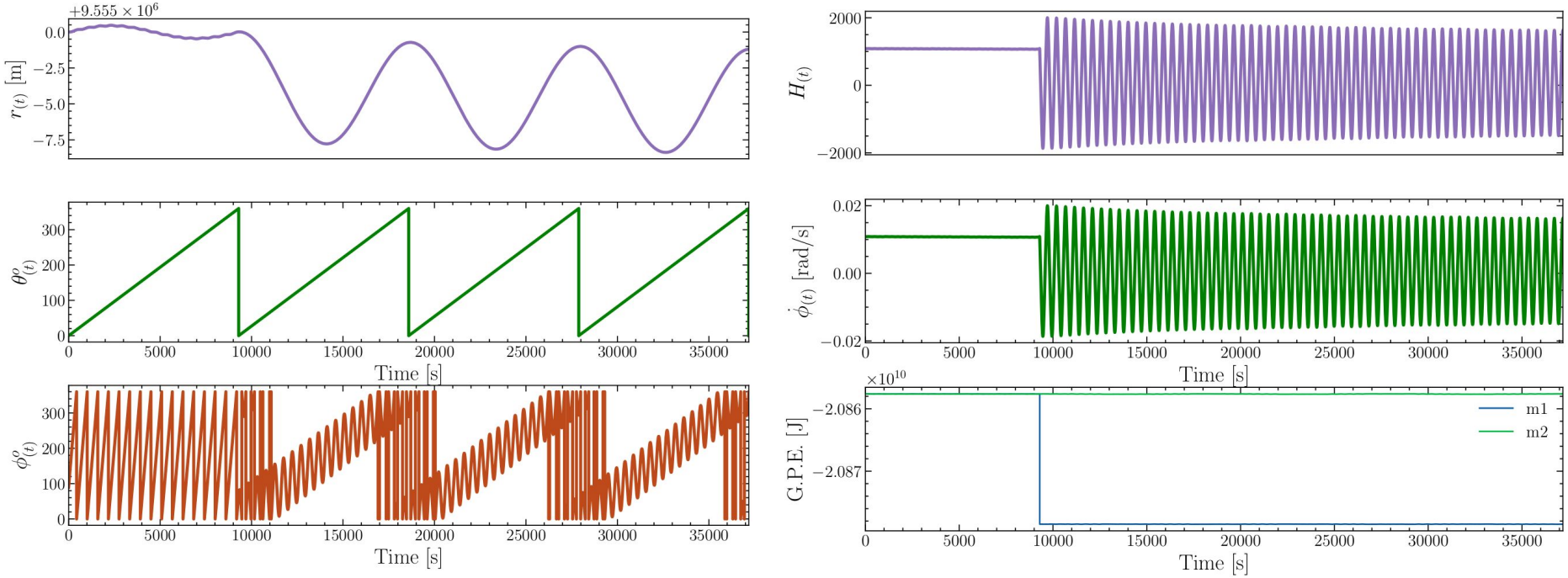
L = 1m

L = 1mm



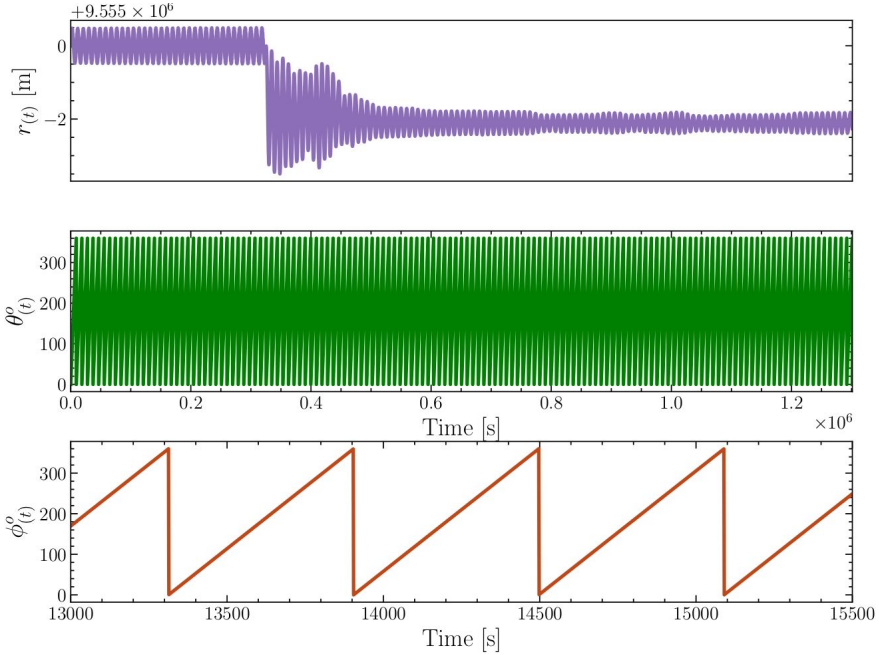


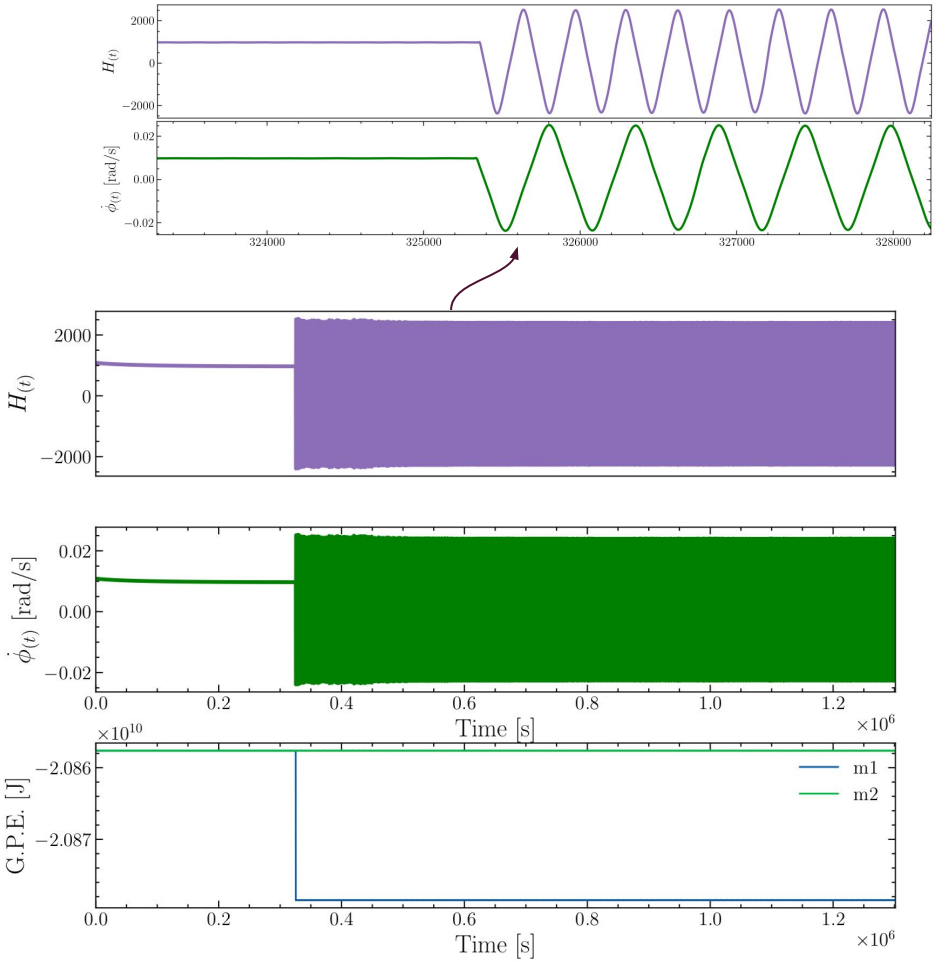
Unbalanced hooking



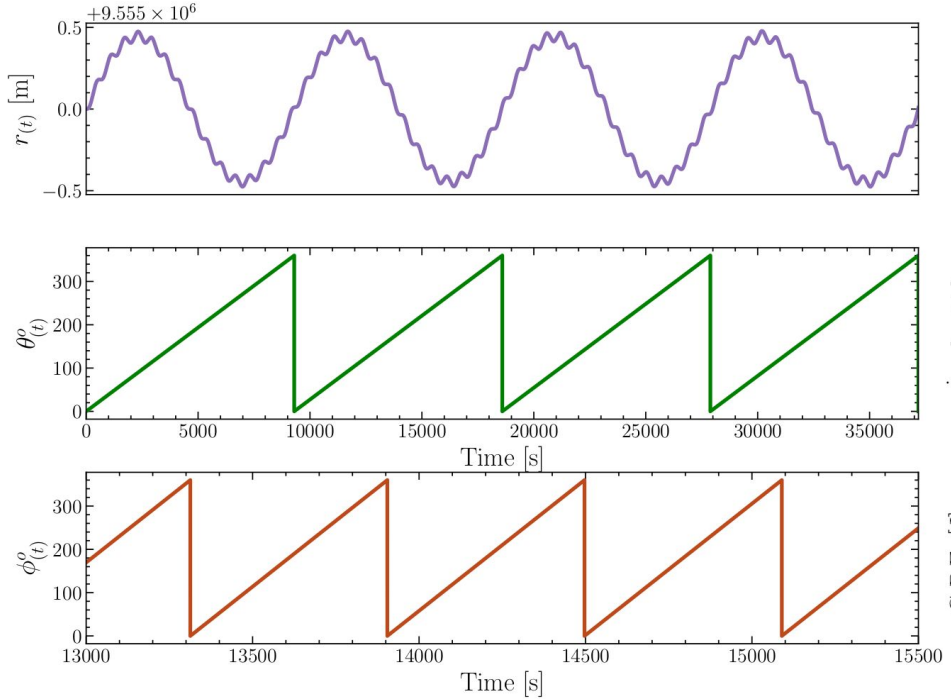
Unbalanced hooking

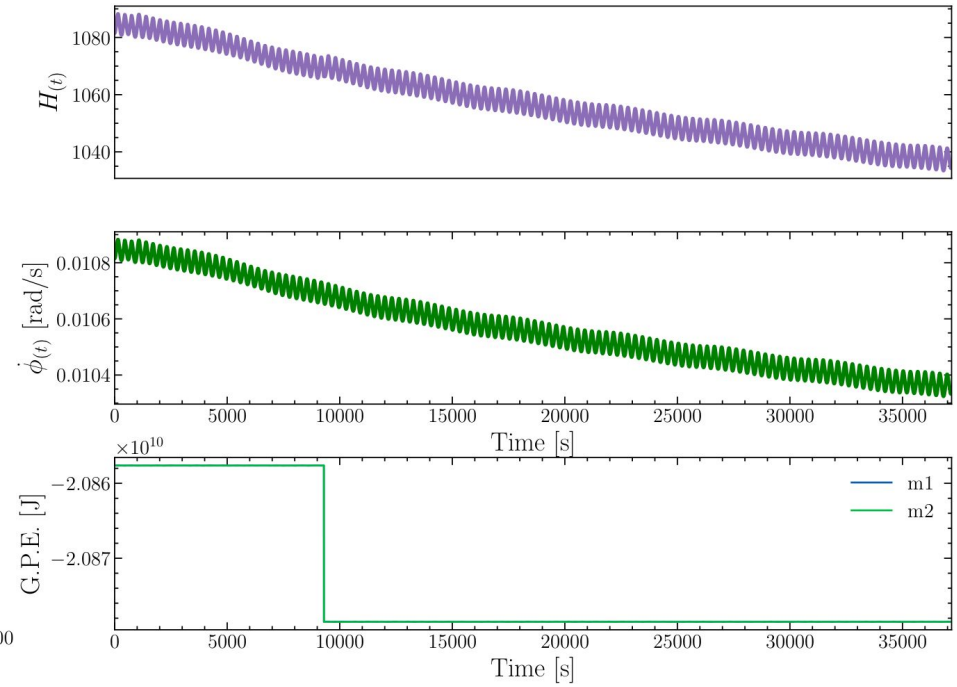
Longer time period



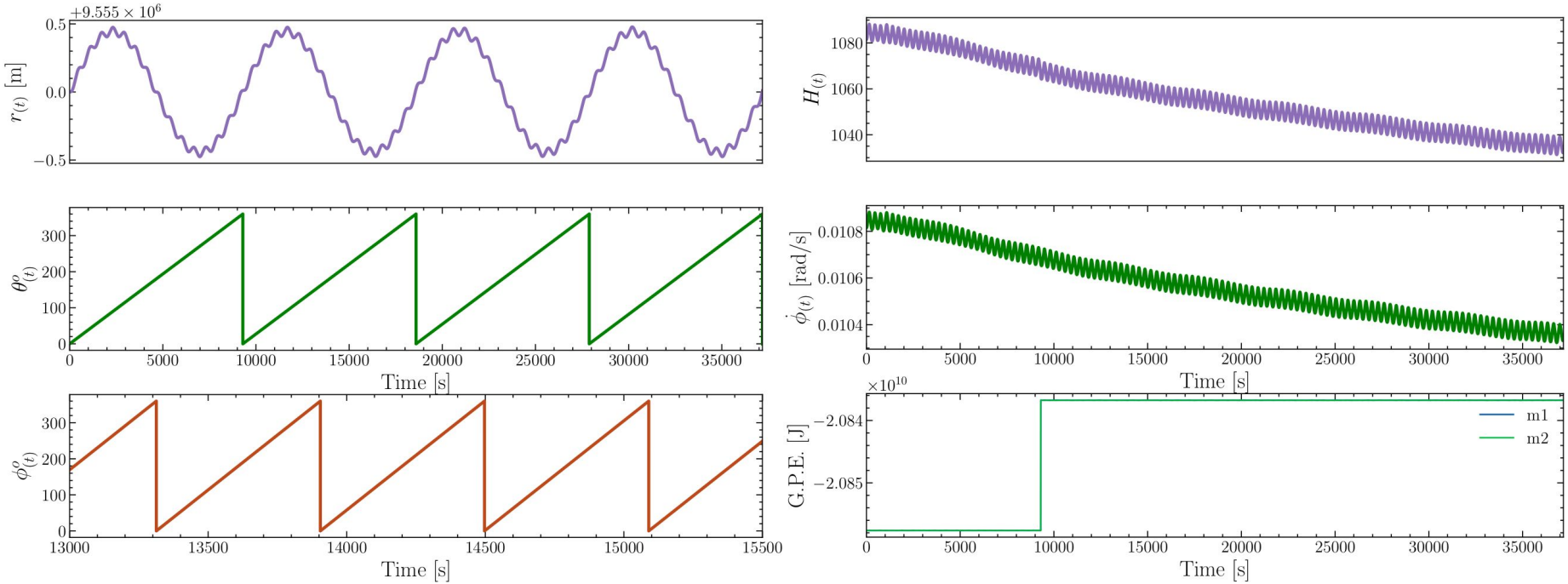


Balanced hooking: Loading phase

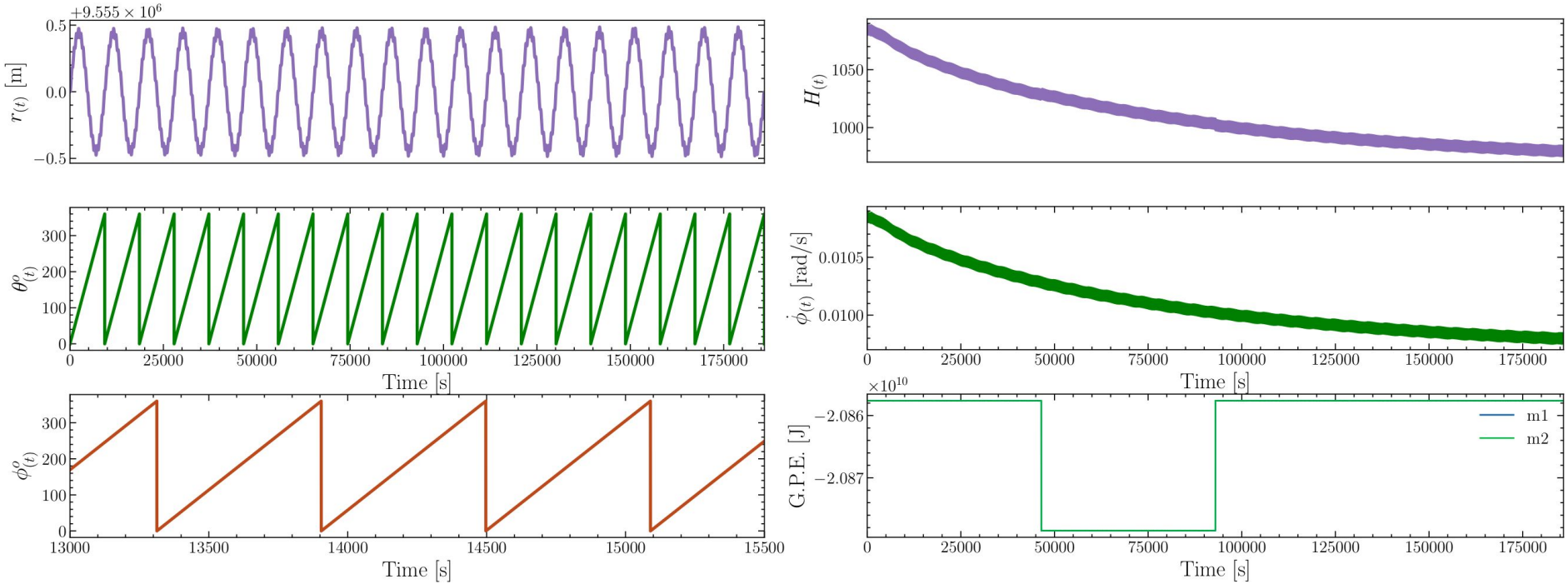




Balanced hooking: Unloading phase



Balanced hooking: Loading + Unloading phase



More to analyse...

- Tether and the tethering
- Dissipative effects
- Scaling effects

YEET

Thank You